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$$f(x,y) = \begin{cases} \dfrac{x^3 - y^3}{x^2 + y^2} & \text{als } (x,y) \neq (0,0) \\ 0 & \text{als } (x,y) = (0,0) \end{cases}$$

Continuïteit:

Stel $x = r\cos\theta$ en $y = r\sin\theta$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{r\to 0} \frac{r^3\cos^3\theta - r^3\sin^3\theta}{r^2\cos^2\theta + r^2\sin^2\theta}$$
$$= \lim_{r\to 0} \frac{r^3(\cos^3\theta - \sin^3\theta)}{r^2} = \lim_{r\to 0} r(\cos^3\theta - \sin^3\theta) = 0$$

$\Rightarrow f$ is continu in $(0,0)$

Partiële afgeleiden:

$$\begin{cases} \dfrac{\partial f}{\partial x}(0,0) = \lim\limits_{\lambda\to 0} \dfrac{f(\lambda,0) - f(0,0)}{\lambda} = \lim\limits_{\lambda\to 0} \dfrac{\frac{\lambda^3}{\lambda^2}}{\lambda} = \lim\limits_{\lambda\to 0} \dfrac{\lambda}{\lambda} = 1 \\ \dfrac{\partial f}{\partial y}(0,0) = \lim\limits_{\lambda\to 0} \dfrac{f(0,\lambda) - f(0,0)}{\lambda} = \lim\limits_{\lambda\to 0} \dfrac{\frac{-\lambda^3}{\lambda^2}}{\lambda} = \lim\limits_{\lambda\to 0} \dfrac{-\lambda}{\lambda} = -1 \end{cases}$$

$\Rightarrow f$ is partieel afleidbaar in $(0,0)$

Afleidbaarheid:

Stel $\overline{h} = (h_1, h_2)$

$$Df(\overline{0}, \overline{h}) = \lim_{\lambda\to 0} \frac{f(\lambda h_1, \lambda h_2) - f(0,0)}{\lambda}$$
$$= \lim_{\lambda\to 0} \frac{\frac{\lambda^3 h_1^3 - \lambda^3 h_2^3}{\lambda^2 h_1^2 + \lambda^2 h_2^2}}{\lambda}$$
$$= \lim_{\lambda\to 0} \frac{\lambda^3(h_1^3 - h_2^3)}{\lambda(\lambda^2 h_1^2 + \lambda^2 h_2^2)}$$
$$= \lim_{\lambda\to 0} \frac{\lambda^3(h_1^3 - h_2^3)}{\lambda^3(h_1^2 + h_2^2)} = \frac{h_1^3 - h_2^3}{h_1^2 + h_2^2}$$

$\Rightarrow f$ is afleidbaar in $(0,0)$

Differentieerbaarheid:

Voor differentieerbaarheid in $(0,0)$ moet gelden: $Df(\overline{0}, \overline{h}) = \begin{pmatrix} \dfrac{\partial f}{\partial x}(0,0) & \dfrac{\partial f}{\partial y}(0,0) \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}$

$$\begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} = h_1 - h_2$$

$$Df(\overline{0}, \overline{h}) = \frac{h_1^3 - h_2^3}{h_1^2 + h_2^2} \neq h_1 - h_2$$

$\Rightarrow f$ is niet differentieerbaar in $(0, 0)$

References